Employee Salary Management

Formulas and attributes, explained from attendance to total salary.

A WORKSPACE GUIDE TO THE CALCULATIONS BEHIND EMS

30-day salary basis

ATTENDANCE

Daily status and paid days

PAYROLL

Earnings and deductions

REPORTS

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A clear path through the EMS calculations

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IMPORTANT SCOPE NOTE

Monthly Incentive source amounts come from the separate Incentive project.
EMS applies employee selection, saved percentage and additional amount; it does not calculate the upstream source-report formulas.

1. Units, month rules and rounding

EMS stores monetary values in integer paise (INR x 100). A displayed value in rupees is paise / 100. Profile, manual earnings, allowances, deductions and advances are input values, not formula outputs. Most percentage/proration calculations round each component separately to a whole paise using decimal ROUND_HALF_UP. Some browser previews use integer Math.round; the production distribution math is kept aligned with the backend calculation.

Term / attribute	Rule used in EMS	Source / note
Salary month	Input must be YYYY-MM; start is first day and end is the real calendar last day.	ems.py: month_bounds
Salary divisor	The active salary profile is forced to 30 days, regardless of calendar month.	ems.py: save_salary_profile
Effective salary profile	Choose the newest profile with effective_month <= selected month.	ems.py / salary_preview
Money conversion	Input INR amount -> int(amount x 100) paise; display -> paise / 100.	ems.py; ems.js
Rounding	For ratio(value, numerator, denominator): round-half-up(value x numerator / denominator) to whole paise.	salary_distribution.py; decimal math JS
Report totals	Sum the displayed numeric values for selected employees; XLSX holds values, not live Excel formulas.	ems.js; ems_workbook.py

2. Attendance attributes feeding salary

Daily attendance for an ERP employee follows this priority: future date = Upcoming; otherwise a recorded punch = Present; otherwise registered leave = Leave; otherwise scheduled weekoff = Week Off; otherwise Absent. A saved EMS daily correction overrides that status for the EMS report only. Manual employee attendance is entered in EMS and does not create shared ERP punches.

Attribute	Formula / mapping	Input or source
Employee / Emp ID / Outlet / Department	Identity-card/account/store fields (or EMS salary display name); no arithmetic.	ERP account, IdentityCard, Store, EMS name override
Daily status	Upcoming / Present / Leave / Week Off / Absent, using the priority above; an EMS correction can override.	Shared punch + leave + historical scheduled weekoff; EMS correction
P, A, CL, WO, L totals	Count selected month days of each status. Half Day contributes 0.5 to P and 0.5 to A.	Daily status; monthly-manual declarations for manual employees
Elapsed days	max(0, min(month end, today) - month start + 1).	Calendar month and current date in IST
Upcoming days	Calendar days in month - elapsed days.	Relevant for open/future salary month
Missing manual attendance	max(0, elapsed days - (P + A + CL + WO + L)). Missing elapsed days are included as absent for pay.	EMSManual monthly totals
Week Off Availed	Count of days with status Week Off in the selected month.	Attendance status and scheduled weekoff history
Week Off Credit	max(0, 4 - Week Off Availed). Credits never go below zero.	Monthly allowance of 4 weekoffs
Payable Days (monthly payroll)	P + CL + WO; half day contributes 0.5 P.	Salary preview pay counts
Unpaid Days (monthly payroll)	A + days before joining + L; half day contributes 0.5 A. CL is always paid; L is unpaid.	Salary preview pay counts
Unpaid Days (IDS distribution)	The monthly salary preview unpaid_days; used as Absent input. If not available, use attendance absent count.	salary-distribution API
Joining date	Identity card joining date; else ERP account registration date; if unavailable, month start. Manual monthly attendance uses month start.	ERP / IdentityCard; EMS manual attendance override
Can Publish	True only if salary setup exists, month ended, no existing payroll, net >= 0, joining <= month end and missing attendance = 0.	Salary preview eligibility checks

For manual monthly attendance, P + A + CL + WO + L may not exceed elapsed days. Incomplete elapsed days remain provisionally unpaid and block publication. Future dates are not treated as absent. Editing EMS attendance updates EMS salary calculations without rewriting source attendance punches.

3. Salary setup and attendance-linked Monthly Payroll

The Salary Setup form displays Expected Basic. It represents the monthly Basic + Allowances total; the browser derives hidden Basic as Expected Basic - entered Allowances. Fixed deductions and account details are entered values. Profile fields become effective from their selected month.

Attribute	Formula / rule	Meaning / caveat
Expected Basic	Monthly Basic + Monthly Allowances.	Displayed salary setup amount; not the same as earned monthly basic after proration.
Monthly Basic / Allowances / Fixed Deductions	Entered amounts stored as paise.	Deductions are not statutory rates; fixed amount is entered by HR.
Week Off Availed / Credit	Attendance count; Credit = max(0, 4 - Availed).	Backend refreshes these when saving setup.
Setup Weekoff Pay	round-half-up((monthly Basic + monthly Allowances) x Week Off Credit / 30).	Preview amount in setup; differs from IDS phase 2 where Total Earning is used.
Setup Incentive	Stored manual setup field.	It is not a calculated value and is not included in the active attendance-linked preview net or current IDS incentive allocation.
Setup Payable preview	Setup Weekoff Pay + Setup Incentive.	Hidden UI preview; not the active published-payroll formula.
Earned Basic	round-half-up(monthly Basic x factor), paise.	Factor below.
Earned Allowances	round-half-up(monthly Allowances x factor), paise.	Rounded separately from Basic.
Proration factor	max(0, 1 - Unpaid Days / 30); force to 0 if all calendar days in the month are unpaid.	Unpaid days include A, pre-joining days and L.
Gross Salary	Earned Basic + Earned Allowances.	Monthly payroll preview only.
Weekoff Pay (monthly payroll)	round-half-up(Gross Salary x Week Off Credit / 30).	Added before fixed deductions.
Incentive / Bonus (monthly payroll)	Sum approved EMS earnings of the matching kind for employee and month; otherwise 0.	Void entries are excluded. Automatic distribution is pending unless using Monthly Incentive/W-O rules.
Net (monthly payroll)	Gross Salary + Weekoff Pay + approved Incentive + approved Bonus - Fixed Deductions.	Monthly attendance-linked preview.
Published bonuses column	Preview Bonus + Preview Weekoff Pay.	Storage combines both; payslip separates them using saved preview snapshot.
Direct/manual payroll net	Entered Basic + Allowances + approved Incentive + approved Bonus - entered Deductions.	Separate legacy payroll endpoint; negative net is rejected.
Paid status	Status changes from published to paid after an external payment is recorded.	EMS does not transfer funds.

4. Salary Distribution and IDS (Phase 1) attribute dictionary

These columns use the category-aware calculation implemented by salary_distribution.py and its matching browser calculation. Amounts below are integer paise until displayed as INR. Let A = capped unpaid days, D = 30 - A, B = effective profile Basic + Allowances, OA = Other Allowances, S = Staff Advance & Deduction, W = Week Off Availed, and Regular = employment category exactly Regular.

Column / attribute	Formula / value rule
Employee ID, Employee Name, Branch, Category	Identity/display fields from ERP/IdentityCard/Store/effective salary category; no salary formula. Branch is omitted from the downloaded IDS sheet.
Absent	A = min(30, max(0, unpaid days)). Includes absences and other unpaid time as defined by attendance rules.
No. of Days	30 (fixed salary divisor).
Working Days	D = 30 - A.
Expected Basic	B = effective profile monthly Basic + monthly Allowances; 0 if no profile.
Actual Basic	round-half-up(B x D / 30).

Column / attribute	Formula / value rule
Other Allowances	Manual monthly input; latest saved value at or before report month carries forward.
Actual VDA	round-half-up(Actual Basic x 625 / 10000) = 6.25% of Actual Basic.
Actual Pay (EPF)	Actual Basic + Actual VDA.
Actual OA	round-half-up(Other Allowances x D / 30).
Total Earning	Actual Pay (EPF) + Actual OA.
Bonus Payable	Proposed = round-half-up(Actual Pay (EPF) x 833 / 10000). Pay proposed amount only when category is Regular AND Total Earning + proposed bonus < INR 21,000; otherwise 0.
Actual Gross Earning	Total Earning + Bonus Payable.
EPF Employee Contribution	Regular: round-half-up(Actual Pay (EPF) x 12 / 100); otherwise 0.
ESIC Employee Contribution	Regular and Actual Gross Earning < INR 21,000: round-half-up(Actual Gross Earning x 75 / 10000); otherwise 0.
Staff Advance & Deduction	Manual amount entered for the selected month; no carry-forward.
Total Deductions	EPF Employee Contribution + ESIC Employee Contribution + Staff Advance.
Net Amount	Actual Gross Earning - Total Deductions. Weekoff pay and Phase 2 incentive are excluded.
Account No. / IFSC Code	Bank details from effective Salary Setup; no calculation.

For Contract or unclassified employees, Bonus Payable, EPF and ESIC are 0. Staff Advance is still deducted. The threshold is strictly below INR 21,000: a result exactly equal to INR 21,000 does not qualify.

5. W/O & Incentive (Phase 2) and Total Salary

Column / attribute	Formula / value rule	Note
Employee ID, Employee Name, Branch, Category	Same identity and effective category fields as Phase 1.	Branch omitted in XLSX phase sheet.
Total W/O	4.	Monthly weekoff allowance.
W/O Availed	Count of Week Off days.	From same month attendance.
W/O Credit	max(0, 4 - W/O Availed).	Never negative.
Gross Salary	Actual Gross Earning from IDS (includes a qualifying Bonus Payable).	Display only; not the basis for weekoff pay.
W/O Credit Pay	round-half-up(Total Earning x W/O Credit / 30).	Uses Total Earning before bonus and deductions.
Incentive	Saved monthly incentive selection result, or manual W/O incentive; a saved manual override takes precedence over calculated monthly amount.	May be 0 if no applicable source amount.
Amount Payable / W/O & Incentive	W/O Credit Pay + Incentive.	Phase 2 payment amount.
Total Salary	IDS Net Amount + Phase 2 Amount Payable.	Only computed when both components are present; otherwise blank.
Total Salary report Absent / Working Days / W/O Credit	Copied from Phase 1 and Phase 2 respective rows.	Not recalculated.

The setup preview may show Weekoff Pay using monthly Basic + Allowances. The active Phase 2 report instead uses IDS Total Earning. The two formulas intentionally differ.

6. Monthly Incentive attributes

EMS applies the latest saved selection/settings effective on or before the selected month. Percent defaults to 100%; additional amount defaults to INR 0. The output is computed from an allowed MonthlyIncentiveRelease report and is not persisted as a calculated amount.

Attribute	Formula / selection rule
Incentive Type	User choice: sales or non-sales; not an arithmetic formula.

Attribute	Formula / selection rule
Sales Outlet / Category Group	Select the source report row by exact (outlet, category group); Base = that exact_summary row total_incentive.
Non-sales Location / Field	Select a non-sales source report field. For Accounts, Ware House, Customer Care, Support or HO: use selected location row value; Central Service uses report totals. Non-central values require an applicable non-null rate. IDS Fund uses selected location IDS fund except MWH. Insufficient-data report returns 0.
Base amount	Matched source report amount; missing/unavailable amount -> 0. EMS does not derive this source amount.
Percentage	Saved input 0-100%, max 2 decimals; stored as basis points (percentage x 100). Latest setting at or before chosen month applies; default 100%.
+ Amount	Manual additional INR amount, saved as paise; effective latest setting carries forward. Default 0.
Exact Incentive	round-half-up(base amount to paise x percentage basis points / 10000) + additional amount in paise. Displayed after conversion to INR.
W/O Incentive output precedence	No allowed monthly release -> no calculated monthly result. Otherwise apply calculated result, then saved manual override replaces it; absent either, use saved manual W/O incentive row.
Manual W/O Incentive	Entered monthly INR amount, saved in paise; if a monthly amount exists (or override already exists), it is saved as a manual override. Reset restores calculation.

The Incentive report itself supplies total_incentive, IDS Fund, non-sales values and rates. Formula logic that produced those external report fields is outside the EMS source files and is not represented as an EMS formula in this document.

7. Attendance Record, manual fields and non-formula attributes

Attribute group	How value is determined	Formula?
Daily employee cells (P, P/2, A, CL, WO, L, AUTO)	HR may enter an EMS correction; AUTO returns to derived shared status. Manual employees may enter EMS-only attendance.	Input / status mapping
Manual P/A/CL/WO/L monthly totals	HR declaration for selected month. Values must be whole nonnegative days; sum <= elapsed days.	Input; incomplete elapsed days are unpaid provisionally
Name / Employee ID / Outlet / department / account details	ERP account, identity card, store, selected salary profile, or separately saved EMS salary name.	Lookup
Effective month / category / Basic / Allowances / Deductions / Other Allowances / Advance	Manager-entered setup or month-specific report input.	Input
Salary status / attendance notes / reviewer / timestamps / audit IDs	Persisted workflow, status and audit metadata.	Workflow/data fields, not formulas
Weekoff schedule	ERP weekly-off setting plus dated history changes.	Lookup/schedule; daily Week Off status is derived from schedule

The legacy EMSAttendanceTotals table is not applied in salary_preview; manual employee declarations are read from EMSManualMonthlyAttendance. The current monthly report computes P/A/CL/WO/L totals from daily states for registered employees.

8. Excel workbook: output values and totals

The downloaded workbook contains IDS, W-O & Incentive, Total Salary, Contract and Regular sheets. It writes the already-calculated values into cells; it does not embed Excel formulas. Branch is omitted from the first two downloaded data sheets. Employee order, filters, account numbers and names are carried from the UI payload.

Workbook field	Calculation / operation
IDS sheet rows	Same Phase 1 fields and formulas listed in Section 4.
W-O & Incentive rows	Same Phase 2 fields and formulas listed in Section 5.
Total Salary row	Net Amount + Amount Payable (computed in Python using Decimal; blank if either source amount is missing).
Contract / Regular rows	Filter Total Salary rows by exact employment category; unknown category is not included in either category sheet.
Total row for each sheet	For configured money columns: round(sum(row value or 0), 2). Totals exclude text/day/count columns.
Currency export	Browser converts paise to INR by dividing by 100. XLSX stores numeric INR amounts with two-decimal format.

Workbook field	Calculation / operation
Month label	YYYY-MM input is formatted as English month abbreviation and year, e.g. 2026-09 -> Sep 2026.

9. Worked example (Regular employee, no absence)

Illustration only; amounts are INR. Assume Expected Basic = 16,000.00; absence/unpaid days = 0; Other Allowances = 0; weekoffs availed = 2; Staff Advance = 0. The input values are treated as whole paise after conversion.

Step	Calculation	Result
Working days	30 - 0	30
Actual Basic	16,000.00 x 30 / 30	16,000.00
VDA	round-half-up(16,000.00 x 6.25%)	1,000.00
Actual Pay (EPF)	16,000.00 + 1,000.00	17,000.00
Total Earning	17,000.00 + 0.00	17,000.00
Proposed Bonus	round-half-up(17,000.00 x 8.33%)	1,416.10
Bonus test	17,000.00 + 1,416.10 < 21,000.00	Pass; Bonus Payable = 1,416.10
Gross	17,000.00 + 1,416.10	18,416.10
EPF	round-half-up(17,000.00 x 12%)	2,040.00
ESIC	round-half-up(18,416.10 x 0.75%)	138.12
Total deductions	2,040.00 + 138.12 + 0	2,178.12
IDS Net Amount	18,416.10 - 2,178.12	16,237.98
Weekoff Credit Pay	round-half-up(17,000.00 x (4-2) / 30)	1,133.33
Phase 2 Amount Payable	1,133.33 + assumed incentive 500.00	1,633.33
Total Salary	16,237.98 + 1,633.33	17,871.31

This example demonstrates why the IDS Net Amount is not itself Total Salary and why the Phase 2 weekoff calculation uses Total Earning rather than Gross Salary.

10. Source map and implementation notes

Project file	Formula/attribute responsibility
ems.py	Attendance summary; profile save; monthly preview and publication; direct payroll; monthly incentive lookup and calculation; salary distribution API inputs; manual monthly attendance validation.
attendance_history.py	Daily attendance status priority and weekoff schedule/history lookup.
salary_distribution.py	Canonical IDS/Salary Distribution formula, including paise rounding and Regular/Contract conditions.
static/salary-distribution-math.js	Browser-side equivalent of the IDS formula for immediate preview.
static/ems.js	Rendered table attributes; staged attendance/input calculation; Phase 2 and Total Salary calculations; workbook payload conversion.
ems_workbook.py	Workbook sheet and column definitions, Total Salary row computation, category sheets and totals.
ems_models.py	EMS stored attributes and constraints; SQLAlchemy columns themselves do not implement formula properties.
EMS_USER_GUIDE.md	User-facing rules and known distinctions between monthly payroll and two-phase distribution.

Change control: This PDF describes the implementation found in the workspace on 06 November 2026. If formulas are changed later, regenerate/update this reference against the new code and rerun the related EMS tests.